Crossed Siamese Vision Graph Neural Network for Remote-Sensing Image Change Detection

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***Abstract*— The development of deep learning in remote-sensing (RS) visual tasks has led to remarkable progress in RS image change detection (CD). However, RS bi-temporal images cover complex and confusing scenes due to natural environmental factors, which presents challenges for CD tasks. How to effectively exploit long-range dependencies and sensitively discriminate real changes with various scales from pseudo-changes are urgent problems. It is especially obvious for the changes in building structures man-made. This article presents a CD approach named CSViG, which utilizes a Siamese vision graph neural network (SViG) with crossed feature fusion. SViG acts as a feature extractor to capture richer short- and long-range dependencies. Crossed feature fusion consists of a horizontal feature fusion module (HFFM) and a vertical feature fusion module (VFFM). HFFM designs a cross-concatenation (CC) way to reveal real changes from pseudo-changes in the same horizontal stage, after which global and local features are extracted by using an attention mechanism and multiscale depth-wise separable convolution (DSConv). VFFM further fuses complementary content from vertical multiple stages to effectively represent change regions of different sizes (tiny or huge) by using an attention mechanism. Extensive comparative experiments conducted on three available building CD datasets demonstrate that the proposed method achieves better CD performance than previous counterparts.**

***Index Terms*— Attention mechanism, change detection (CD), deep learning, feature fusion, graph neural networks, remote-sensing (RS).**

I. Introduction

With the rapid development of remote-sensing (RS) technology and the convenient accessibility of massive RS image datasets, change detection (CD) has become one of the most attractive tasks in RS visual research. The purpose of CD is to recognize changes between bi-temporal images of the same area captured at different times, assigning the label of change or unchange to each pixel of the area. The detection of change in RS has significant practical applications in land cover surveys, [1], [2], [3], urbanization monitoring [4], [5], and natural disaster evaluation [6]. Finding effective approaches to leverage long-range dependencies [7] and accurately distinguish real changes at different scales from pseudo-changes [7], [8], [9] are pressing challenges that need to be addressed. In RS image processing, short-range dependencies refer to the relationships between adjacent pixels in an image, while long-range dependencies [7] refer to the relationships between nonadjacent pixels that may be far apart in the image. Capturing both short- and long-range dependencies is important for many RS applications. The concept of pseudo-changes [9] refers to unchanged areas, which can be easily misjudged visually as having changed due to differences in natural conditions (seasonal shift, weather change, light intensity, and so on) when capturing bi-temporal images of the area at different times. Efficient discrimination between real and pseudo-changes is also essential for CD.

Based on the research objects, early traditional CD methods can be divided into two categories: pixel-based CD methods [10] and object-based CD methods [11], [12]. Some algorithms with the strengths of succinct feature representation and rapid change extraction, such as change vector analysis (CVA) [13] and principal component analysis (PCA) [14], have been employed extensively. These methods commonly analyze changes of interest by exploiting the rich spectral, textural, structural, and geometric information in RS bi-temporal images. However, due to natural factors, such as weather, light, and seasonal shifts, the RS bi-temporal images acquired at different times contain random noise and pseudo-changes, which cause formidable challenges for CD tasks. The traditional CD methods that seriously rely on manually extracted low-level visual image features are insufficient to detect critical change information.

In the last decade, deep learning has achieved outstanding performance in computer vision since it possesses effective and reliable feature representation capability. More scholars have applied deep learning to RS visual tasks, such as image scene classification [15], semantic segmentation [16], road extraction [17], and CD [7] and have made enormous progress. Convolutional neural networks (CNNs) with shift invariance and localization characteristics are utilized to extract semantic features of images as the basis for downstream tasks. Numerous CD approaches draw on the full convolutional

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network (FCN) [18], [19], [20] to build end-to-end frameworks for revealing changes of interest, they focus only on short-range dependencies but ignore long-range dependencies which are also critical to distinguishing regions of change. Some recent methodologies enlarge reception field (RF) by applying dilated convolution [21], [22] or establish global information by introducing attention mechanisms [9], [23] to improve the performance of detection models, but they still have room for improvement to relate long-range dependencies in space and time. To address the above long-range context issue, Transformer is introduced as an encoder for modeling the context within semantic tokens that represent high-level concepts of changed areas [7], [24]. Nevertheless, their computation is notably inefficient, and the presence of redundant semantic tokens may diminish the performance.

Graph neural networks (GNNs) [25], [26], [27] have been implemented in various fields due to their ability to efficiently process unstructured and unordered data, and their applicability scenarios are being constantly broadened in the development and evolution process. Han et al. [28] introduced a new vision GNN (ViG) architecture, the image was split into some patches that can be regarded as nodes of graph structure, and they then created edges between adjacent nodes. Inspired by ViG [28], we apply Siamese ViG (SViG) as an encoder in the CD task. Graphs are suitable for representing the dependencies and interrelationships between various entities. Our intuition is that high-level concepts of changes can be represented as nodes in the graph, and dependencies between various changes (both short- and long-range) can be connected by edges.

Unlike general image-processing tasks, CD input data typically consists of pre- and postchange images, so how to effectively and concisely fuse the features of bi-temporal images is vital. One simple solution [29] to generate differential features is by directly subtracting postchange features from prechange features, but this approach may overlook areas where changes are not obvious. Another scheme [30] concatenates the pre- and postchange features into new features and then yields the differential features by applying CNNs, yet the redundant features following the concatenation might weaken the feature representations of changed regions. To further amplify the gap between changed and unchanged differential feature representations, attention mechanisms [23], [29], [30] are adopted to reinforce the interaction of global and local feature information. Therefore, we design a new horizontal feature fusion module (HFFM) that involves cross-concatenation (CC) way discovering difference between changed and unchanged areas (or pseudo-changes as shown in Fig. 1(a), channel-space attention (CSA) capturing global context, and multiscale depth-wise separable convolution (DSConv) focusing on local information. In general, the low-stage features contain massive low-level texture information, while the high-stage features provide abundant semantic content. Inspired by FPN [31] and AFF [32], we adopt a vertical feature fusion module (VFFM), which similarly utilizes the attention mechanism, to fuse multistage refined differential features output by HFFM. VFFM not only facilitates the detection of tiny changed objects [as shown in Fig. 1(b)], but also improves the segmentation boundary of changed areas with various scales.

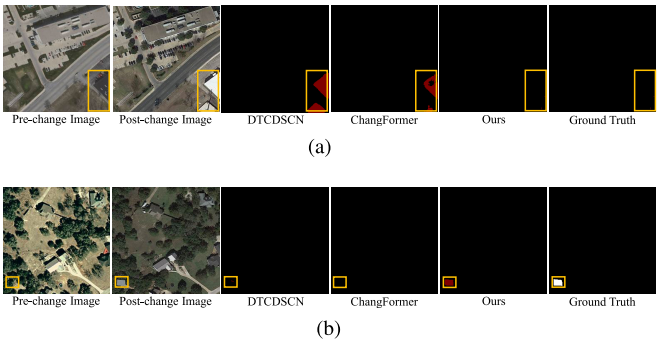


Fig. 1. Challenges in CD tasks. (a) Bi-temporal images contain numerous pseudo-changes due to natural conditions. (b) Detecting a large number of tiny changed areas can be difficult. Each row presents a comparison of the visualization results for a single case. Our proposed HFFM and VFFM are designed to address these issues.

In this article, we propose a novel CD approach named CSViG, which is based on SViG and crossed feature fusion that refers to HFFM in the horizontal direction and VFFM in the vertical direction. We first employ SViG with shared parameters as the encoder to individually extract features containing affluent short- and long-range dependencies of the bi-temporal images at different stages [(1/4), (1/8), (1/16), and (1/32) of input image size]. We can model high-level change concepts as nodes in a graph and capture the dependencies between different changes through edges. HFFM then combines the pre- and postchange feature maps of the same horizontal stage to output a set of differential features. This process is accomplished through three key components: CC, which identifies differences between changed and unchanged areas (or pseudo-changes); CSA, which captures global context; and multiscale DSConv, which focuses on local information. HFFM is capable of accurately identifying real-change areas while effectively suppressing pseudo-change objects. Finally, we utilize VFFM to fuse multistage refined differential features to obtain the ultimate change prediction, which is also effective for tiny changed areas of interest. The findings illustrated in Fig. 1 provide evidence that the incorporation of these elements can effectively mitigate the challenges associated with CD and enhance the performance of the model. It is worth mentioning that, while attention mechanisms are utilized in many visual tasks, the modified attention mechanisms used in VFFM and HFFM are specifically tailored to address the unique challenges of CD tasks.

In summary, the main contributions of this work are as follows.

1) We propose HFFM to effectively discover differences between changed objects and unchanged (or pseudo-changed) regions, which is achieved by using the CC way and fusing local information and global context.
2) We introduce VFFM to combine refined differential features from multiple stages, which enables improved representation and detection of changed objects with various scales (tiny or huge).

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3) We validate the effectiveness of our method on three publicly accessible CD datasets, and the experiments show that our proposed approach achieves state-of-the-art CD performance.

The rest of this article is organized as follows: Section II introduces related works on CD in detail. The proposed CD method is described in Section III. The experimental results and the discussion are shown in Sections IV and V, respectively. Finally, Section VI concludes this work.

II. Related Work

The CD is an essential task in RS, which has been extensively explored, and many new technologies have been developed by researchers in recent years. The main CD methods can be categorized into traditional CD methods and deep-learning CD methods. In this section, we briefly introduce the relevant works of CD from traditional to neural networks in RS images.

A. Traditional CD Methods

Early traditional CD methods are commonly based on machine-learning algorithms utilizing algebra, transformation, and classification. Some traditional CD approaches [10], [13], [33], [34] directly compare spectral and texture information of pixels in bi-temporal RS images to generate the differential maps to obtain the final change result. Ferraris et al. [13] applied CVA that conducted difference calculations on data of each band from different periods to exploit the intrinsic characteristics of bi-temporal images. Saux and Randrianarivo [35] proposed the change-index histogram of oriented gradients (CI-HOG) to capture local statistics of change indices. A new postclassification method with iterative slow feature analysis (ISFA) and Bayesian soft fusion (BSF) is proposed to obtain reliable and accurate CD maps in [34]. Although the above approaches are simple to implement, they ignore the spatial background information and generate noise during processing to seriously weaken the detection performance. Other methods [12], [36], [37] use feature transformation algorithms to obtain object information from RS bi-temporal images. Zhang et al. [38] introduced a novel object-based CD technique incorporating multiscale uncertainty analysis. PCA was also used to enhance the change information from stacked multisensor data in [14]. However, these traditional CD methods depend heavily on low-level information from RS bi-temporal images that are susceptible to natural conditions, such as weather, light, and season. They can achieve effective results in simple scenarios, but the final predictions of the model are usually not accurate enough in complex and shifty scenarios.

B. Deep-Learning CD Methods

In recent years, due to the rapid development of RS technology, adequate high-resolution RS bi-temporal images are easily available, and more researchers have begun to explore the application of deep learning in RS visual tasks. CD methods based on deep learning are also being explored and developed gradually. Below, we review the mainstream deep-learning CD methods based on two different classification criteria.

1) Different Fusion Stage: According to different fusion stages of bi-temporal information, main methods can be roughly divided into image-level [39], [40], [41], [42], [43] and feature-level [5], [9], [18], [29], [30], [44], [45], [46], [47]. Image-level fusion methods simply concatenate raw RS bi-temporal images with the same number of channels into a new image with a double number of channels to meet the requirement of a single input to the segmentation network. A novel end-to-end CD method [40] was presented based on the effective UNet++ [48] architecture that employs dense skip connections to achieve multiscale feature extraction, to capture subtle real changes in challenging scenes, but it requires a large number of true change maps and limits the widespread use to a certain extent. Peng et al. [42] proposed a dense attention method consisting of several upsampling attention units to simultaneously capture change information in both spatial and channel dimensions. Although these image-level fusion methods can achieve end-to-end CD [43] simply and effectively, the deep features of individual original images fail to be extracted by the network to help distinguish the boundaries of change region, which leads to the final result being unsatisfactory. Feature-level fusion methods are also explored in more articles, which normally employ Siamese networks [18], [45], [46], [47] with shared parameters as feature extractors to handle RS bi-temporal images, respectively. Daudt et al. [18] presented FC-Siam-Conc and FC-Siam-Di fusing the feature maps via concatenation and difference before skipping connections, respectively, yet they failed to consider the global content due to the limited RF of the network. Currently, more models use attention mechanisms to capture significant global information. MapsNet [29] applied a multiattention module (MAM) and pixel-shuffle image fusion network (PSIFN) to aggregate multilevel contextual information and implement feature fusion, but it was not sensitive to small areas of change. Zhang et al. [30] utilized CBAM [49] to fuse multilevel deep features of raw images, then they introduced change map losses of intermediate layers to achieve end-to-end network training, and a shortcoming was that using concatenation to fuse pre- and postchange features at the same level, which contained plenty of disturbing information. These Siamese networks usually have more accurate predictions owed to feature fusion modules, so we choose a feature-level fusion approach with an attention mechanism as the baseline in this article.

2) Different Encoder Structures: Most CD networks primarily consist of an encoder and a decoder. The encoder is chiefly designed to extract features, while the decoder recognizes regions of change. According to different encoder structures, they can be categorized as CNN-based [9], [18], [22], [50], [51], [52], [53], Transformer-based [7], [24], [54], and others [55], [56]. The CNN extracts high-level semantic features of each temporal image to reveal the change of interest. Zhang et al. [22] used dilated convolution expanding the receptive field to focus on more change regions. ESCNet [50] incorporated the feature extraction capability of the CNN

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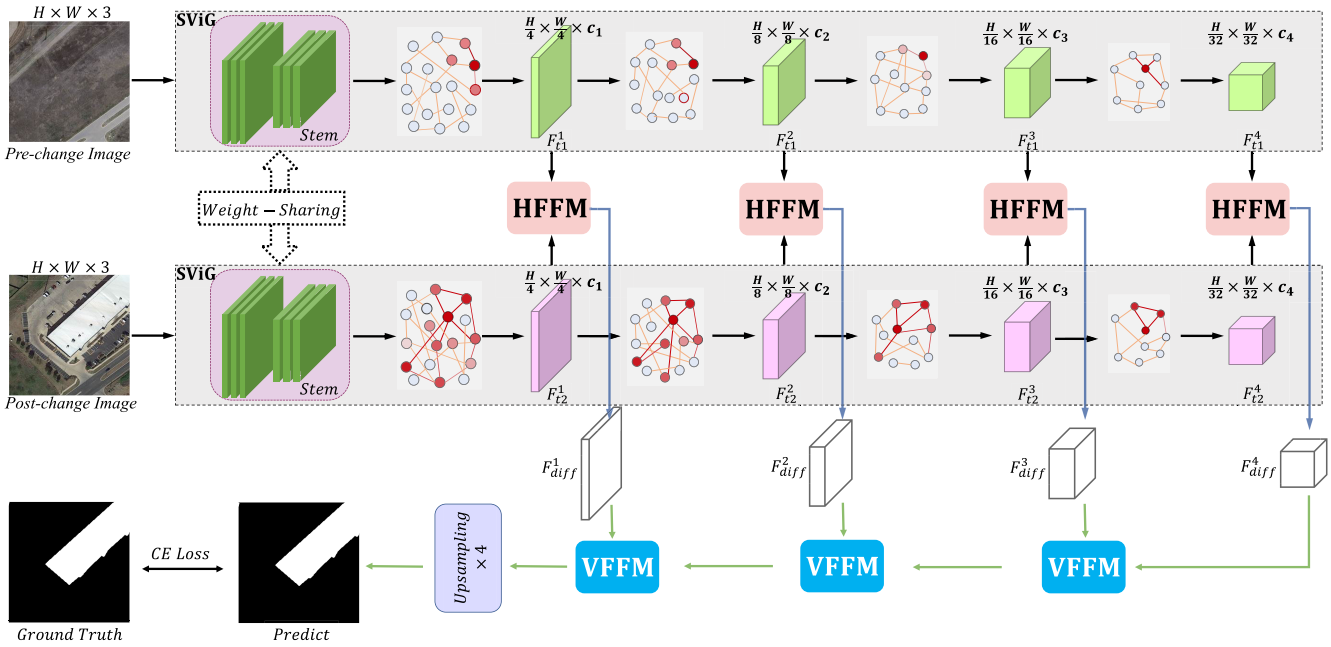


Fig. 2. Overview of the proposed CD method CSViG. First, SViG extracts multistage features from pre- and postchange images, respectively. Then, HFFM fuses bi-temporal features of the same horizontal stage to get a set of differential features. Finally, VFFM further fuses complementary content from vertical multiple stages. The specific structure of HFFM and VFFM are presented in Figs. 4 and 5, respectively.

and the edge-smoothing effect of superpixel segmentation to improve performance. Other CNN-based [9], [51], [52] approaches introduced attention mechanisms to capture global relationships and information among pixels in space–time. Due to the ease of handling long-range dependencies, Transformer has gained more attention in the field of RS image processing recently. Chen et al. [7] considered that the change of interest could be described by a few high-level semantic tokens. They used a Transformer encoder [57] to model contexts in the compact token-based space–time. ChangeFormer [58] unified hierarchically structured Transformer encoder with a multilayer perceptron (MLP) [59] decoder to efficiently render multiscale long-range details required for accurate CD. Meanwhile, generative adversarial networks (GANs) [60], [61] and recurrent neural networks (RNNs) [62] are also adopted to detect the change of interest. Zhao et al. [61] proposed MeGAN automatically to explore seasonal invariant features for pseudo-change suppressing and real CD. However, CNN-based methods fail to effectively capture long-range dependencies in RS images, while redundant semantic tokens weaken the performance of the model in Transformer-based methods. To some extent, the ViG [28] can be effective in mitigating the issues mentioned above.

Therefore, our proposed CSViG is well-suited for the task of CD in RS images. To begin with, we utilize SViG as the encoder to individually extract features that contain short- and long-range dependencies of the bi-temporal images at different stages. While CNN-based methods [9], [18], [50], [51], [52] with limited RF can capture local information, they may overlook valuable global content that could be beneficial. Although the Transformer-based approaches [24], [54], [57], [58] can also build long-range relationships to focus on all content, the redundant tokens may weaken the performance of the model at a certain level. CSViG connects the most relevant nodes (high-level concepts of changed or unchanged regions) to preserve vital features and remove redundant information. In addition, our proposed model chooses feature-level fusion way as the baseline. Currently, the popular ways of obtaining differential features in feature-level fusion CD methods [18], [45], [46], [47] are limited in discriminating between real changes and pseudo-changes, since they use simple operations that are subtraction, absolute of subtraction, and concatenation. Our designed CC way is capable of identifying the differences between changed and unchanged areas effectively and automatically. Meanwhile, the attention mechanisms employed in VFFM and HFFM have been customized and modified to effectively tackle the distinctive challenges posed by CD tasks.

III. Proposed Approach

In this section, a comprehensive overview of the proposed network is presented, followed by a detailed description of each individual module comprising the network.

A. Overview

The overall architecture of our proposed CD method CSViG is illustrated in Fig. 2. It is composed of three components: an SViG encoder, an HFFM, and a VFFM. To fully harness long-range dependencies and enhance our model's ability to extract high-quality feature representations, we construct graph structures of bi-temporal images individually being input into SViG for automatically extracting multistage features. HFFM then obtains differential features based on the feature maps of pre- and postchange at the same horizontal stage, which actively learns the changes between the bi-temporal images.

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Moreover, VFFM is applied to capture more meaningful information by using an attention mechanism that allows multistage refined differential feature maps to interact with adjacent features.

For the data of the CD task, $\mathcal{X}_{t1}$ and $\mathcal{X}_{t2}$ represent a pair of images in the same area at different times $t1$ and $t2$, respectively, and $\mathcal{Y}$ indicates the corresponding label with change annotations. We describe the proposed method specifically using notations as follows.

1) At first, the RS bi-temporal images $\mathcal{X}_{t1}$ and $\mathcal{X}_{t2}$ are converted into graphs, with nodes representing changed or unchanged regions and edges indicating dependencies (short- and long-range) between various regions. The graphs are input into the weight-sharing SViG that can reason about dependencies and downsample features to extract a set of multistage feature maps $\mathcal{F}_m = \{F_m^1, F_m^2, F_m^3, F_m^4\}$, $m = t1, t2$.
2) Next, HFFM admits pre- and postchange feature maps from the same horizontal stage to compute their differential feature. CC way adaptively calculates pre- and postchange features of the same stage to get the coarse basic differential features, CSA and multiscale DSConv compute the optimal features fusion scheme from both global and local directions. We can obtain a group of multistage refined differential features $\mathcal{F}_{\text{diff}} = \{F_{\text{diff}}^1, F_{\text{diff}}^2, F_{\text{diff}}^3, F_{\text{diff}}^4\}$.
3) Then, VFFM allows for further interaction and fusion of refined differential features that contain different content in adjacent phases. It leverages these multistage refined differential features to obtain intermediate change maps $\mathcal{C} = \{C^1, C^2, C^3\}$.
4) Finally, the change map C^1 is upsampled to generate the final change prediction $\mathcal{Y}_p$, so it and the corresponding label $\mathcal{Y}$ compute the cross-entropy loss $\mathcal{L}$.

B. Siamese ViG

The encoder employs an SViG with two weight-sharing branches to extract features from the pre- and postchange images individually, which calls for us to obtain the graph structure of the bi-temporal image pairs. For an image X with the size of $H \times W \times 3$, we divided it into N patches. Each patch is transformed into a feature vector $x_i \in \mathbb{R}^D$, and we can produce a group of vectors $X = [x_1, x_2, \ldots, x_N]$, in which D is the feature dimension and $i = 1, 2, \ldots, N$. These feature vectors can be considered as unordered nodes which are denoted as $\mathcal{V} = \{v_1, v_2, \ldots, v_N\}$ in a graph. As with ViG [28], we construct edges between each node v_i and its $k = 9$ nearest neighbor nodes $\mathcal{N}(v_i)$ to obtain the set of all edges $\mathcal{E} = e_1 \cup e_2 \cup \cdots \cup e_i \cup \cdots \cup e_N$, where e_i denotes the edge set between node v_i and $\mathcal{N}(v_i)$. In this way, we obtain a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Our intuition is that high-level concepts of changes can be represented as nodes in the graph, and dependencies between various changes (both short- and long-range) can be connected by edges, reasoning about dependencies between regions of change by GNNs.

In general, GNNs consist of two main operations: aggregation and update of features. The aggregation operation computes the representation of a node by aggregating features of neighbor nodes, and the update operation further merges the aggregated feature by using a fully connected layer. The RS image $\mathcal{X}_m$ can be constructed as $\mathcal{G}_m$, $m = t1, t2$, that is the input feature, and the module can be expressed as

$$\mathcal{Z}_m = \sigma(\texttt{VIG}(\mathcal{X}_m, W_{\text{in}}))W_{\text{out}} + \mathcal{X}_m \tag{1}$$

where VIG denotes operation of ViG in [28], $\mathcal{X}_m \in \mathbb{R}^{N \times D}$ is the node feature, both W_{in} and W_{out} are learnable parameters, and σ is the activation function GeLU. The long-range dependencies between nonadjacent nodes (patches) in the RS image that are far apart but connected by edges may be captured by VIG, while the CNN is difficult to handle. Since we mainly refer to ViG [28], we briefly describe the process here, without describing the specific aggregation and update operations, refer to ViG [28] for more details. After that, we change the number of channels (c_1, c_2, c_3, c_4) and size [(1/4), (1/8), (1/16), (1/32) of the original input image size] of the features using convolution and downsampling operations to obtain the multistage features of RS bi-temporal images $\mathcal{F}_m = \{F_m^1, F_m^2, F_m^3, F_m^4\}$, $m = t1, t2$ (as shown in Fig. 2). The SViG encoder can capture both short- and long-range dependencies to help the model obtain more excellent feature representations.

C. Horizontal Feature Fusion Module

The target of the CD task is to detect regions of change in RS bi-temporal images, so learning the difference between pre- and postchange features is essential. To effectively uncover the difference between changed and unchanged regions as well as to widen the gap in their feature representations, HFFM is designed to obtain differential features by comparing features of pre- and postchange at the same horizontal stage. It first uses CC operation to obtain basic differential features and then utilizes attention mechanisms and multiscale DSConv in parallel to further improve the representation ability of differential features. The following describes the implementation of this module.

The most popular ways of obtaining differential features in feature-level fusion CD methods are presented in Fig. 3: (a) subtraction, (b) absolute of subtraction, and (c) concatenation. Although they are simple to operate, they are limited in discriminating between real changes and pseudo-changes. Therefore, CC is designed to calculate the basic differential features. Since we adopt a weight-sharing feature extractor to process the bi-temporal images individually, information on features corresponding to the same channel of pre- and postchange is considered to be correlated. Unlike the concatenation way, combining two channels corresponding to the same position in the pre- and postchange features into a channel feature group, we similarly get a new mixed feature with a double number of channels. Then, DSConv is adopted to learn the difference of bi-temporal images and filter out more pseudo-change features. The process of calculating the basic differential features can be expressed by the following equation:

$$B_{\text{diff}}^s = \texttt{Conv}\big(\texttt{BN}\big(\texttt{ReLU}\big(\texttt{DSConv}\big(\texttt{CC}\big(F_{t1}^s, F_{t2}^s\big)\big)\big)\big)\big) \tag{2}$$

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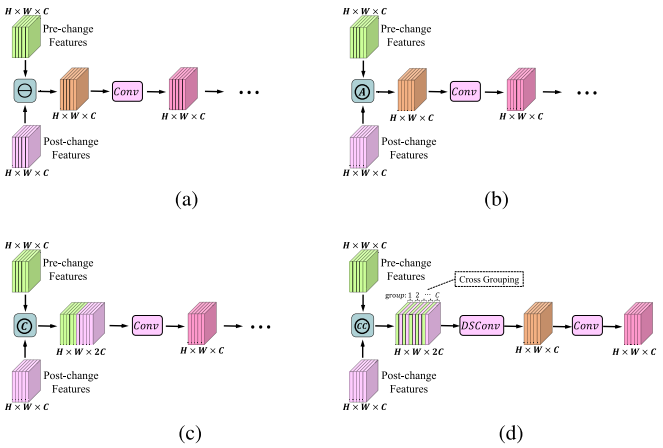


Fig. 3. Main ways of generating basic differential features. (a) Subtraction of two feature maps. (b) Absolute after subtraction of two feature maps. (c) Concatenation of two feature maps. (d) CC of two feature maps designed in CSViG.

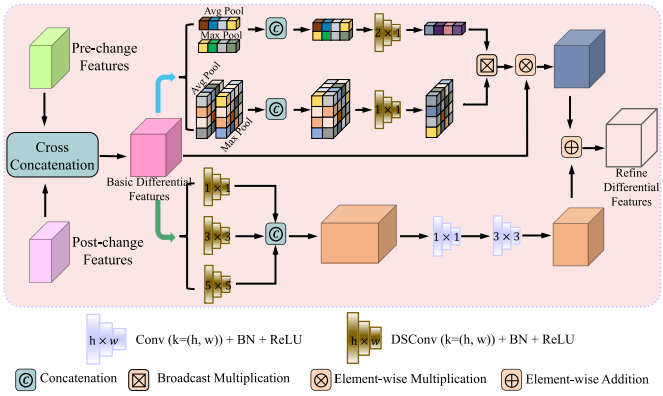


Fig. 4. Overview of HFFM. First, CC is designed to obtain the basic differential features. Then, refined differential features are gained using the attention mechanism and the multiscale DSConv.

where F_{t1}^s and F_{t2}^s denote the features of pre- and postchange images from the sth hierarchical stage, $s = 1, 2, 3, 4$, CC denotes that the tensor is cross-concatenated, and BN is the batch normalization. Note that both Conv and DSConv use a kernel with the filter size of 3×3 by default, and the specific convolution kernel size with $h \times w$ is represented in (3).

Expanding the gap between changed and unchanged features to improve the representation capability, we handle the basic differential features in parallel by applying the attention mechanism to capture ample global information and multiscale convolution to focus on adequate local information. As illustrated in Fig. 4, the blue arrow points to the CSA module. The single basic differential feature $B \in \mathbb{R}^{D\times H\times W}$ is produced by the CC way, where $H \times W$ is the size of feature B, and D means the number of channels. Computation of the channel attention map M_F^c and spatial attention map M_F^s are detailed as follows:

$$M_F^c = \mathtt{ReLU}\left(\mathtt{Conv}^{2\times 1}\left(\left[\mathtt{AvgP}^c(B); \mathtt{MaxP}^c(B)\right]\right)\right) \quad (3)$$

$$M_F^s = \mathtt{ReLU}\left(\mathtt{Conv}^{7\times 7}\left(\left[\mathtt{AvgP}^s(B); \mathtt{MaxP}^s(B)\right]\right)\right). \quad (4)$$

First, the input feature map is squeezed through an average pooling operation and a max-pooling operation both on the spatial and channel axes. Then, two feature vectors are concatenated to compose a new feature map, and we use convolution to compute the global channel and spatial context. Finally, a ReLU function is attached to assign the context of each channel and space to get M_F^c with the size of $D \times 1 \times 1$ and M_F^s with the size of $1 \times H \times W$. The eventual channel-spatial feature map holding more global information B_G is calculated as follows:

$$B_G = \mathtt{Sigmoid}\left(M_F^c \boxtimes M_F^s\right) \otimes B \quad (5)$$

where $\boxtimes$ denotes broadcast multiplication, which performs element-wise operations between arrays or tensors that have different sizes or shapes, without the need for explicit looping constructs (broadcast addition of (13) operates similarly). $\otimes$ denotes element-wise multiplication, and $B_G \in \mathbb{R}^{D\times H\times W}$.

The green arrow points to the multiscale convolution capturing adequate local information in Fig. 4. To reduce the parameters, we concatenate multiple feature maps that are produced DSConv with various sizes and then fuse them and reduce the number of channels using point-wise convolution (PWConv) with the size of 1×1. Finally, local information can be expressed with the following equation:

$$O = \left[\mathtt{DSConv}^{1\times 1}(B); \mathtt{DSConv}^{3\times 3}(B); \mathtt{DSConv}^{5\times 5}(B)\right] \quad (6)$$

$$B_L = \mathtt{ReLU}\left(\mathtt{Conv}^{3\times 3}\left(\mathtt{ReLU}\left(\mathtt{Conv}^{1\times 1}(O)\right)\right)\right). \quad (7)$$

BN layer also is omitted. Once the plentiful global content B_G and local information B_L are obtained by the attention mechanism and multiscale convolution, we aggregate them using the easiest and directest way to calculate refined differential feature $F_{\text{diff}} \in \mathbb{R}^{D\times H\times W}$

$$F_{\text{diff}} = B_G \oplus B_L \quad (8)$$

where $\oplus$ denotes element-wise addition and $F_{\text{diff}} \in \mathbb{R}^{D\times H\times W}$ denotes refined differential feature.

The above describes the process of computing individual refined differential features from the same horizontal stage pre- and postchange images. In general, multistage features of the bi-temporal images are attained in the encoding step. We can obtain a group of multistage refined differential features $\mathcal{F}_{\text{diff}} = \{F_{\text{diff}}^1, F_{\text{diff}}^2, F_{\text{diff}}^3, F_{\text{diff}}^4\}$, in which $F_{\text{diff}}^s \in \mathbb{R}^{D_s\times H_s\times W_s}$, and $s = 1, 2, 3, 4$.

D. Vertical Feature Fusion Module

A CD model with excellent performance should not only be sensitive and effective in discriminating between real changes and pseudo-changes, but also be able to detect changed objects with different scales and edge details of the change. Inspired by AFF [32], VFFM is introduced to obtain intermediate change maps by fusing refined differential features from adjacent vertical stages.

Fig. 5 presents the overall VFFM. For the input low-stage feature maps $F_l \in \mathbb{R}^{D_l\times H_l\times W_l}$ and high-stage feature maps $F_h \in \mathbb{R}^{D_h\times H_h\times W_h}$, they are first turned into features with the same number of channels D_m via PWConv. Then, the high-stage features are upsampled to possess the same size as the low-stage features, and the features from two adjacent

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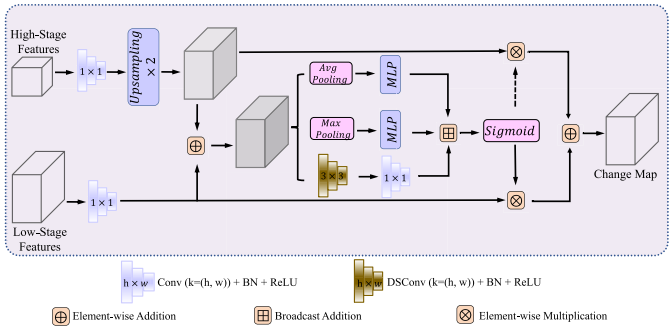


Fig. 5. Overview of VFFM. A change map is obtained by fusing refined differential features of adjacent vertical stages.

vertical stages are summed to obtain their mixed features as follows:

$$F_h^{\text{up}} = \texttt{Up}\big(\texttt{Conv}^{1\times1}(F_h)\big)$$
$$F_l^m = \texttt{Conv}^{1\times1}(F_l)$$
$$F_{l\text{-}h} = F_l^m \oplus F_h^{\text{up}} \tag{9}$$

where $\oplus$ denotes element-wise addition, Up means upsampling operation (bi-linear interpolation), $F_h^{\text{up}} \in \mathbb{R}^{D_m\times H_l\times W_l}$, and $F_{l\text{-}h} \in \mathbb{R}^{D_m\times H_l\times W_l}$. $F_h^{\text{up}} \in \mathbb{R}^{D_m\times H_l\times W_l}$ is obtained by changing the number of channels via convolution and changing the size via upsampling. Then, we further aggregate the global and local contexts of mixed features to generate a weight matrix, which enables a more optimal fusion of features in adjacent stages to improve the performance of the model. The global context is likewise computed using an average pooling operation and a max-pooling operation in the spatial axis, after which the simple MLP is applied to further transform the global information, as shown in the following equation:

$$F_a = \texttt{MLP}(\texttt{AvgP}^{\texttt{c}}(F_{l\text{-}h})) \tag{10}$$
$$F_m = \texttt{MLP}(\texttt{MaxP}^{\texttt{c}}(F_{l\text{-}h})). \tag{11}$$

To maintain our model as lightweight as possible, we capture local content just using DSConv and PWConv

$$F_p = \texttt{ReLU}\big(\texttt{Conv}^{1\times1}\big(\texttt{ReLU}\big(\texttt{DSConv}^{3\times3}(F_{l\text{-}h})\big)\big)\big). \tag{12}$$

When the global and local contents are output, we assemble them to yield the weight matrix

$$W = \texttt{Sigmoid}\big(F_a \boxplus F_m \boxplus F_p\big) \tag{13}$$

where F_a with the size of $D_s \times 1 \times 1$ and F_m with the size of $D_s \times 1 \times 1$ are global contents, F_a with the size of $D_s \times H_l \times W_l$ is local content, and $\boxplus$ denotes the broadcast addition. Finally, change map C is computed as follows:

$$C = \big(F_l^m \otimes W\big) \oplus \big(F_h^{\text{up}} \otimes (1 - W)\big). \tag{14}$$

To gradually fuse the set of multistage refined differential features $\mathcal{F}_{\text{diff}}$ obtained from the HFFM, we first get the change map from the two adjacent feature maps of the highest stage by VFFM. Subsequently, the change map and lower refined differential feature to obtain a new change map. Eventually, the set of change maps $\mathcal{C} = \{C^1, C^2, C^3\}$ are generated by repeatedly using VFFM.

TABLE I
EXPERIMENTAL PARAMETER SETTINGS FOR DIFFERENT CD METHODS

Method	batch size	initial lr	optimize	epochs
FC-EF [18]	16	0.001	Adam	200
FC-Siam-Di [18]	16	0.001	Adam	200
FC-Siam-Conc [18]	16	0.001	Adam	200
DTCDSCN [5]	16	0.001	Adam	100
IFNet [30]	16	0.0001	Adam	200
SNUNet [47]	8	0.001	Adam	100
BIT [7]	8	0.01	SGD	200
ChangeFormer [58]	8	0.0001	AdamW	200
ICIFNet [63]	8	0.01	SGD	200
DMINet [64]	8	0.01	SGD	200
CSViG	8	0.0001	AdamW	200

E. Loss Calculation

By leveraging all the modules above, the final change map is one-fourth the size of the original input image, which can be restored to the same size as the original image using upsampling (bi-linear interpolation) to output the change prediction $\mathcal{Y}_p$ with a size of $H \times W \times 2$. The CD is considered a per-pixel binary classification task, and thus, we minimize the cross-entropy loss to optimize network parameters for this task. In other words, we train the network to classify each pixel in the input image as either changed or unchanged by minimizing the difference between predicted and ground-truth labels using cross-entropy loss, the formula is as follows:

$$\mathcal{L} = -\frac{1}{WH}\sum_{w=1}^{W}\sum_{h=1}^{H}\sum_{c=1}^{C}\mathcal{Y}^{w,h,c}\cdot\texttt{log}\mathcal{Y}_p{}^{w,h,c} \tag{15}$$

where W and H are the width and height of the remote image, respectively, $C = 2$ means the number of change and unchange categories to be identified, $\mathcal{Y}^{w,h,c}$ denotes the label that the pixel in coordinates w and h belongs category class c according to the ground truth, and $\mathcal{Y}_p^{w,h,c}$ represents the prediction probability output by our model for each pixel position of the input bi-temporal images toward category c.

IV. EXPERIMENTS

In this section, we first introduce the experimental environment and provide some parameter settings. Then, the evaluation metrics and comparative methods are briefly described. Finally, we analyze the quantitative data and qualitative visualizations from all experiments conducted on three publicly available CD datasets to demonstrate the validity of our proposed model.

A. Experimental Setup

Our method is implemented on PyTorch and trained using a single NVIDIA Titan XP graphical processing unit (GPU). We randomly initialize the network, and the AdamW optimizer with weight decay equal to 0.01 and beta values equal to (0.9, 0.999) is applied to optimize its parameters. The initial learning rate is set as 0.0001 and linearly decays to 0 until trained for 200 epochs. We define the value of batch size as 8. At the end of each training epoch, we perform validation to

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TABLE II
COMPARISON OF INFORMATION FROM THE THREE USED CD DATASETS

Datasets	Image-size	Image-modality	Space-resolution	Image-pairs	Train / Val / Test	Image-type	Changing-objects
LEVIR-CD [51]	1024×1024	RGB	0.5 m/pixel	637	7120 / 1024 / 2048	Satellite images	building
DSIFN-CD [30]	512×512	RGB	-	3988	14400 / 1360 / 192	Satellite images	building
WHU-CD [53]	32207×15354	RGB	0.2 m/pixel	1	5947 / 743 / 744	Aerial images	building

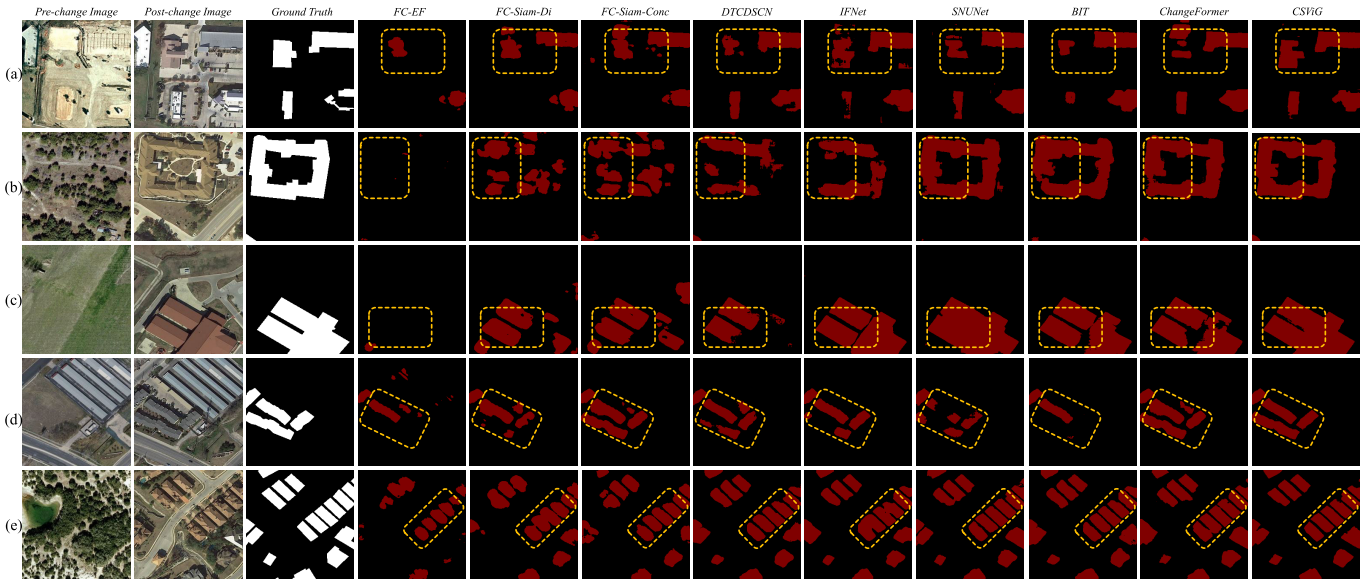


Fig. 6. Qualitative visualizations of different CD methods on *LEVIR-CD*. (a)–(e) represent 5 samples of test set. From left to right, the first two columns are pre- and postchange images, respectively. The third column represents ground truth, and the others are prediction results of different CD methods.

gain a dynamic view of the prediction performance. Finally, we use the best model from the validation set to evaluate the test set. Note that all CD approaches fail to use pretrained and are executed on the same platform using the same system, so all methods are fairly compared. The values for experimental parameters of the other comparison methods are set according to the respective articles, and the specific details are shown in Table I.

To increase the diversity of training data, we employ a range of popular data augmentations: image random rotation (90°, 180°, and 270°); image random flipping (horizontal and vertical); image random blurring; image color jitter.

B. Datasets

To verify the usefulness of our proposed method, we use three publicly available RS image CD datasets: LEVIR-CD [51], DSIFN-CD [30], and WHU-CD [53]. Important information about these three CD datasets is presented in Table II. All datasets are high-resolution image pairs, which are all visible light-based RGB images, and they all contain only building changes. We describe them separately below.

1) LEVIR-CD: The LEVIR-CD [51] is a public large-scale building CD dataset that contains 637 very high-resolution (VHR, 0.5m/pixel) RS image pairs with a size of 1024 × 1024 pixels. These bi-temporal images show significant land-use change covering a variety of building types such as villa residences, tall apartments, small garages, large warehouses, and so on. Due to the limitation of GPU memory size and the convenience of comparison with other CD methods, we crop all original images of the default dataset split (training/validation/test) into nonoverlapping patches of size 256 × 256. Eventually, 7120/1024/2048 samples of RS bi-temporal are obtained for training/validation/test, respectively.

2) DSIFN-CD: The DSIFN-CD [30] is a challenging general CD dataset collected from Google Earth. It contains high-resolution satellite bi-temporal images with changes in different land-cover objects of six cities in China. This dataset provides 3600/340/48 image pairs of default size 512 × 512 for default training/validation/test set, respectively. To maintain consistency in data size for all our experiments, we similarly crop these images into nonoverlapping patches of size 256 × 256 in the three sets. Therefore, 14 400/1360/192 samples are obtained for training/validation/test of our experiment, respectively.

3) WHU-CD: The WHU-CD [53] primarily covers some areas that are rebuilt in the following years after an earthquake occurred. This dataset contains bi-temporal pairs consisting of aerial images captured in 2012 and 2016, respectively. Since it includes aerial images of 32 507 × 15 354 pixels, we also crop all very high-resolution (VHR, 0.2m/pixel) pairs into small and nonoverlapping patches of size 256 × 256. Finally, we form 6096/762/762 RS image pairs for training/validation/test, respectively.

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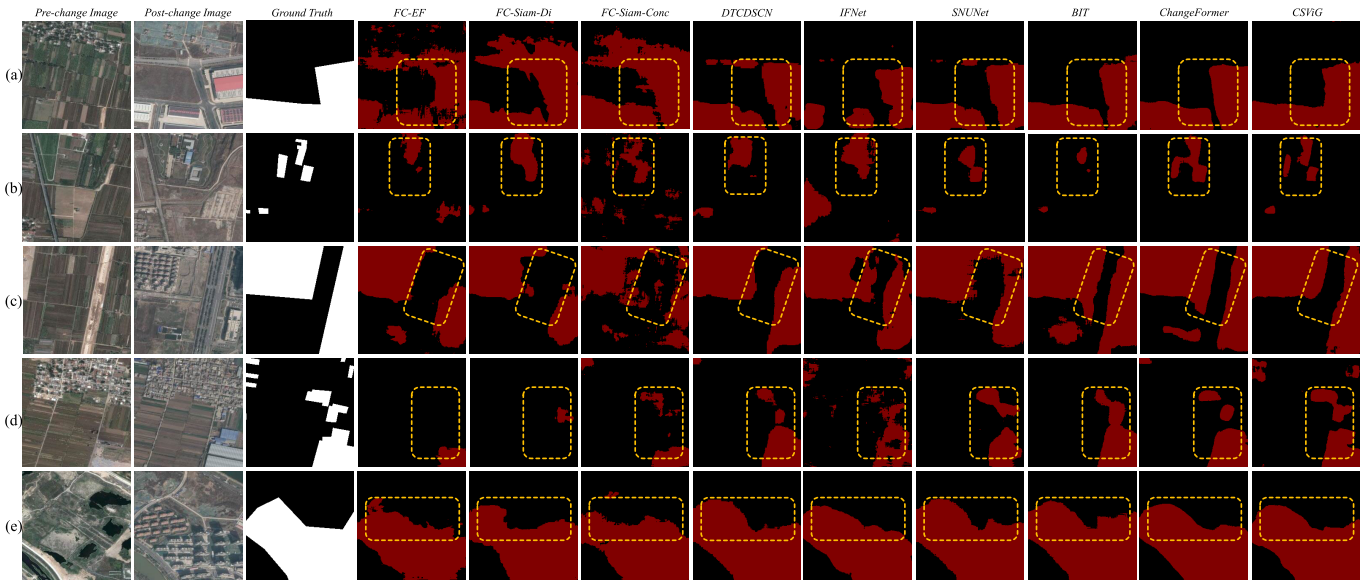


Fig. 7. Qualitative visualizations of different CD methods on *DSIFN-CD*. (a)–(e) represent 5 samples of test set. From left to right, the first two columns are pre- and postchange images, respectively. The third column represents ground truth, and the others are prediction results of different CD methods.

C. Evaluation Metrics and Comparative Methods

1) Evaluation Metrics: To evaluate the performance of our proposed method, we report intersection over union (IoU) and $F1$ scores about the change category as the primary evaluation indices. Furthermore, precision (P), recall (R) of the change category, and overall accuracy (OA) are also measured. They are expressed as follows:

$$\begin{aligned} \text{IoU} &= \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} \\ F1 &= \frac{2}{P^{-1} + R^{-1}} = \frac{2PR}{P + R} \\ P &= \frac{\text{TP}}{\text{TP} + \text{FP}} \\ R &= \frac{\text{TP}}{\text{TP} + \text{FN}} \\ \text{OA} &= \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FN} + \text{FP}} \end{aligned} \tag{16}$$

where TP is the number of true positives, FP is the number of false positives, TN is the number of true negatives, and FN is the number of false negatives.

2) Comparative Methods:

a) FC-EF [18]: The FC-EF concatenates the pre- and postchange images into a single whole, which is fed into the Unet structure with skip connections.

b) FC-Siam-Di [18]: The FC-Siam-Di calculates absolute values between bi-temporal feature maps extracted by weigh-sharing encoding layers to detect change regions.

c) FC-Siam-Conc [18]: The FC-Siam-Conc concatenates pre- and postchange feature maps output by the same encoder, after which the multiscale differential feature maps are concatenated to fuse features.

d) DTCDSCN [5]: The DTCFSCN utilizes channel attention and spatial attention to exploit more discriminative features for better detection of changes in interest.

e) STANet [51]: The STANet devises a self-attention mechanism to exploit the spatial–temporal relationship to detect the finer details of changing objects.

f) IFNet [30]: The IFNet applies an attention mechanism to fuse multilevel deep features of bi-temporal images and introduces change map losses in intermediate layers of the network to further improve detection performance.

g) SNUNet [47]: The SNUNet uses a densely connected Siamese network to extract high-level features and then channel attention and deep supervision are applied to enhance the discriminative capability of features.

h) BIT [7]: The BIT considers change regions as high-level semantic tokens and uses an encoder–decoder-based Transformer to model context information.

i) ChangeFormer [58]: The ChangeFormer unified hierarchically structured Transformer encoder with an MLP decoder to capture the long-range dependencies and improve detection accuracy.

j) ICIFNet [63]: The ICIFNet extracted global features and local features using the CNN and Transformer, respectively, after which the Conv attention module allowed features of the same size extracted by different backbone to interact with each other, improving the feature representation and performance of the model.

k) DMINet [64]: The DMINet proposed an intertemporal joint-attention (JointAtt) block to steer the global feature distribution, prompting information coupling between inner representations while suppressing task-irrelevant interference.

D. Result and Visualization on LEVIR-CD

Table III reports overall comparison results on the test set of LEVIR-CD [51]. The first-best, second-best, and third-best experimental data are marked in red, blue, and black with bold font, respectively. Note that all CD approaches fail to use pretrained and are executed on the same platform using the same system, so all methods are fairly compared.

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TABLE III

AVERAGE QUANTITATIVE RESULTS OF DIFFERENT CD METHODS ON *LEVIR-CD*. ALL VALUES ARE REPORTED IN PERCENTAGE (%). COLOR CONVENTION: **BEST** (RED), **SECOND-BEST** (BLUE), AND **THIRD-BEST** (BOLD)

Method	P	R	$F1$	IoU	OA
FC-EF [18]	81.38	66.48	73.18	57.70	97.52
FC-Siam-Di [18]	85.43	75.85	80.35	67.16	98.11
FC-Siam-Conc [18]	83.85	82.82	83.33	71.42	98.31
DTCDSCN [5]	91.12	87.79	89.43	80.87	98.94
STANet [51]	83.81	**91.00**	87.26	77.40	98.66
IFNet [30]	91.64	86.72	89.11	80.36	98.90
SNUNet [47]	91.41	88.22	89.79	81.47	98.98
BIT [7]	**92.33**	87.19	89.69	81.30	98.97
ChangeFormer [58]	91.12	88.64	89.86	81.59	98.98
ICIFNet [63]	91.32	88.64	**89.96**	**81.75**	**98.99**
DMINet [64]	**92.52**	**89.95**	**90.75**	**82.99**	**99.07**
CSViG	**92.35**	**90.71**	**91.52**	**84.37**	**99.14**

TABLE IV

AVERAGE QUANTITATIVE RESULTS OF DIFFERENT CD METHODS ON *DSIFN-CD*. ALL VALUES ARE REPORTED IN PERCENTAGE (%). COLOR CONVENTION: **BEST** (RED), **SECOND-BEST** (BLUE), AND **THIRD-BEST** (BOLD)

Method	P	R	$F1$	IoU	OA
FC-EF [18]	55.79	73.49	63.42	45.43	85.59
FC-Siam-Di [18]	**69.48**	47.67	56.55	39.42	87.55
FC-Siam-Conc [18]	45.11	**75.81**	56.56	39.43	80.21
DTCDSCN [5]	50.04	74.93	60.01	42.86	83.03
STANet [51]	**67.71**	61.68	**64.56**	**47.66**	**88.49**
IFNet [30]	39.92	**76.05**	52.36	35.47	76.48
SNUNet [47]	49.80	70.08	58.57	41.41	82.91
BIT [7]	**68.97**	57.31	62.60	45.56	**88.36**
ChangeFormer [58]	50.88	**76.06**	60.97	43.86	83.45
ICIFNet [63]	63.72	65.42	63.77	46.85	87.32
DMINet [64]	60.97	70.92	**65.81**	**48.75**	87.41
CSViG	62.88	70.55	**66.50**	**49.81**	**88.92**

TABLE V

AVERAGE QUANTITATIVE RESULTS OF DIFFERENT CD METHODS ON *WHU-CD*. ALL VALUES ARE REPORTED IN PERCENTAGE (%). COLOR CONVENTION: **BEST** (RED), **SECOND-BEST** (BLUE), AND **THIRD-BEST** (BOLD)

Method	P	R	$F1$	IoU	OA
FC-EF [18]	71.63	67.25	69.37	53.11	97.61
FC-Siam-Di [18]	47.33	77.66	58.81	41.66	95.63
FC-Siam-Conc [18]	60.88	73.58	66.63	49.95	97.04
DTCDSCN [5]	92.57	85.34	**88.81**	**79.87**	**99.15**
STANet [51]	79.37	85.50	82.32	69.95	98.52
IFNet [30]	87.92	81.33	84.50	73.16	98.82
SNUNet [47]	77.05	65.23	70.64	54.61	97.85
BIT [7]	85.07	**90.70**	87.79	78.24	**98.99**
ChangeFormer [58]	89.08	80.08	84.34	72.92	98.81
ICIFNet [63]	**92.98**	85.56	88.32	79.24	98.96
DMINet [64]	**93.84**	**86.25**	**88.69**	**79.68**	98.97
CSViG	**94.69**	**86.79**	**90.57**	**82.76**	**99.28**

As can be observed from the quantitative results, our proposed model consistently manifests significant superiority over other approaches. In other words, the proposed network achieves the best CD performance in terms of $F1$, IoU, P, and OA metrics. More specifically, the $F1$/IoU/P/R/OA score exceeds ICIFNet by 1.56/2.62/1.03/2.07/0.15% and outperforms ChangeFormer by 1.66/2.78/1.23/2.07/0.16%. In addition, the $F1$/IoU/R/OA score exceeds the recent method DMINet by 0.77/1.38/0.76/0.07%.

To directly compare the detection performance of different methods, the qualitative visualization of our proposed method compared with other CD methods on test images from LEVIR-CD is illustrated in Fig. 6, and the areas with significant dissimilarity between the results of different methods are marked by orange rectangles. We can observe that our method accomplishes superior prediction than other compared methods. Concretely, most other approaches struggle to effectively detect the edge texture information of complex changed buildings, while our approach can reveal them sensitively, such as Fig. 6(a), (c), and (e). Meanwhile, our model is completely and effectively aware of the change regions in Fig. 6(b) and (d). In other words, our model predicts fewer false-positive samples and false-negative samples. Hence, both the quantitative experimental results and the qualitative visualizations adequately illustrate the superior performance of our proposed CD method.

E. Result and Visualization on DSIFN-CD

From the experimental data about the DSIFN-CD [30] dataset in Table IV, the performance of our method is greater than other CD methods. Specifically, our method achieves the highest accuracy 66.50/49.81/88.92% than the other comparative nine methods in terms of $F1$/IoU/OA evaluation metrics. Similarly, all models do not use the pretrained weight of other datasets to ensure a fair comparison.

For a more intuitive assessment, we also visualize some representative prediction results of the test set as shown in Fig. 7. Our method not only successfully detects large change objects, but also pays close attention to finer details, as illustrated in Fig. 7(a), (c), and (d). For some small changing regions, such as Fig. 7(b) and (d), our method can discover them as well, but other compared methods fail to work well. This can also adequately explain our model's effectiveness for changed objects with different scales.

F. Result and Visualization on WHU-CD

The experimental results of the WHU-CD [53] dataset are presented in Table V. The data indicate that our method reaches excellent outcomes in the five evaluation metrics mentioned. Furthermore, the $F1$/IoU/P/R/OA score outperforms DMINet by 1.88/3.08/0.85/0.54/0.31%.

Likewise, Fig. 8 presents five examples of the prediction visualization on the test set of WHU-CD. Since shadows of buildings in Fig. 8(b) are close to the color of the buildings themselves, the other methods also consider the shadows as areas of change, but our method precisely distinguishes the buildings from shadows. In contrast, the shadows of higher buildings are projected on the lower building in Fig. 8(e), and the compared methods regard the changed buildings as unchanged regions, but our model identifies them well.

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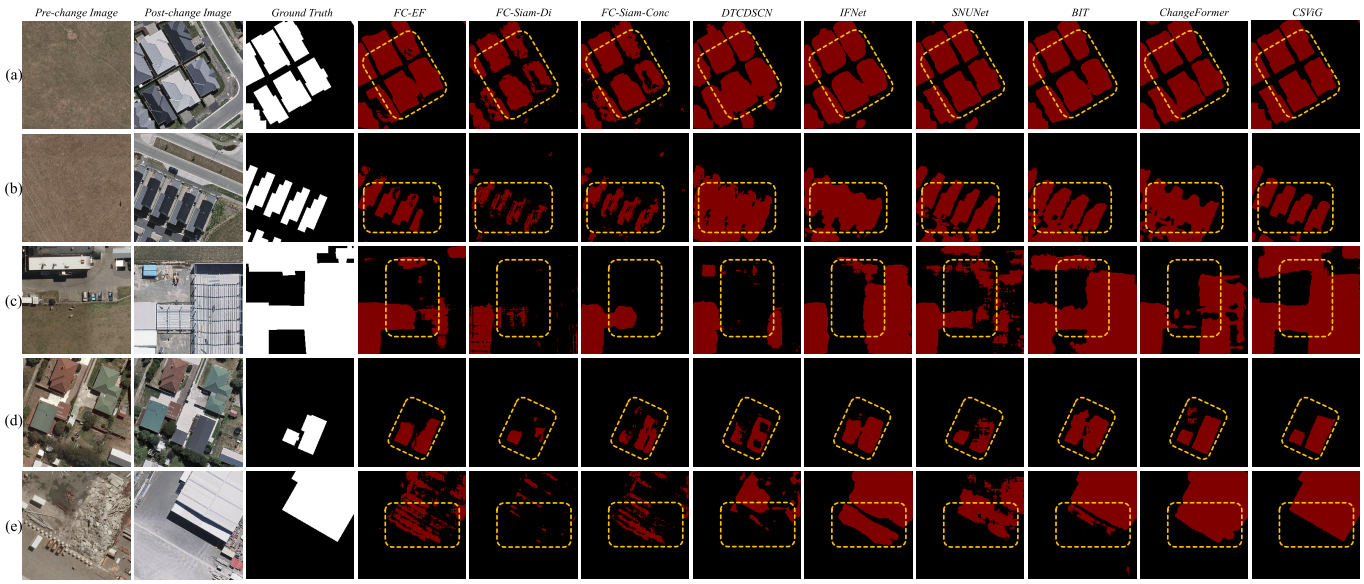


Fig. 8. Qualitative visualizations of different CD methods on *WHU-CD*. (a)–(e) represent 5 samples of test set. From left to right, the first two columns are pre- and postchange images, respectively. The third column represents ground truth, and the others are prediction results of different CD methods.

In addition, for the effect of the light factor, our proposed method is also better at detecting the change region of interest in Fig. 8(c) and (d). To sum up, our model is more robust and can effectively distinguish the real changes and pseudo-changes.

V. Discussion

A. Ablation Study

Our model uses the weight-sharing SViG as an encoder to focus on more long-range dependencies, which boosts the discriminative capability of features. The HFFM fuses pre- and postchange features at the same horizontal stage and the VFFM fuses multistage differential features vertically. In this section, we conduct comparative experiments to further validate the effectiveness of different components in our proposed method. Here, all experiments are done on LEVIR-CD to show the performance.

1) SViG Encoder Emphasizes Short- and Long-Range Dependencies: First, SViG splits the RS image into a group of patches that are used to build a graph structure to automatically focus on short- and long-range dependences. In other words, the local and global information in RS bi-temporal images can be effectively captured to improve model representation. To better explain the effectiveness of SViG, we visualize the feature heat maps of it and other methods at multiple stages in Fig. 9, which shows the comparative results of four different samples in LEVIR-CD. Compared with other CD methods, the extractor used by us obtains feature maps at the lower stages, such as Fig. 9(a) and (b), containing more abundant and specific information about global and local details, and feature maps at the higher stages, such as Fig. 9(c) and (d), revealing more semantic information about changed or unchanged objects. More specifically, the ChangeFormer [58] in Fig. 9(1)(d) and the SNUNet [47] in Fig. 9(2)(d) fail to focus effectively on complex areas of change, and unchanged areas that may degrade model performance are highlighted in the DTCDSCN [5] of Fig. 9(3)(d). The visualization results visually illustrate that the SViG encoder focuses on more useful local and global feature content. The feature information extracted by SViG is more reliable and comprehensive, which means that the encoder has better feature extraction ability. This is attributed to the fact that SViG focuses on both short- and long-range dependencies and reasons for the relationship between regions of change.

TABLE VI

Experimental Results When CC and Other Ways Are Used to Acquire Differential Features. The Values of Evaluation Metrics Are Reported as Percentage (%)

Ways	Params	P	R	$F1$	IoU	OA
Subtraction	hwC^2	91.55	90.29	91.06	83.60	99.10
Absolute-Value	hwC^2	91.74	88.96	90.25	82.23	99.04
Concatenation	$2hwC^2$	91.23	90.44	90.89	83.30	99.08
Cross-Concatenation	$hwC(C+2)$	**92.35**	**90.71**	**91.52**	**84.37**	**99.14**

2) CC Is Useful to Fuse Bi-Temporal Features: The characteristic of the CD task is the input bi-temporal images. The experiments are done to compare pre- and postchange feature fusion methods, including subtraction, absolution of subtraction, concatenation, and CC. As seen in Table VI, we analyze their parametric quantities (refer to Fig. 3 for the specific structure) in the column named "Params," where w and h denote the size of convolution kernel, and C denotes the number of feature channels ($C \gg h, w$). In addition, we also compare $F1$/IoU/OA performance when using them separately on the LEVIR-CD dataset. The experimental data suggest that our way achieves higher scores on evaluation metrics. We believe that CC can reliably and effectively identify real-changed features and suppress pseudo-changed features.

3) SViG, HFFM, and VFFM Can Improve Model Performance: In Fig. 9 presented earlier, a qualitative analysis is

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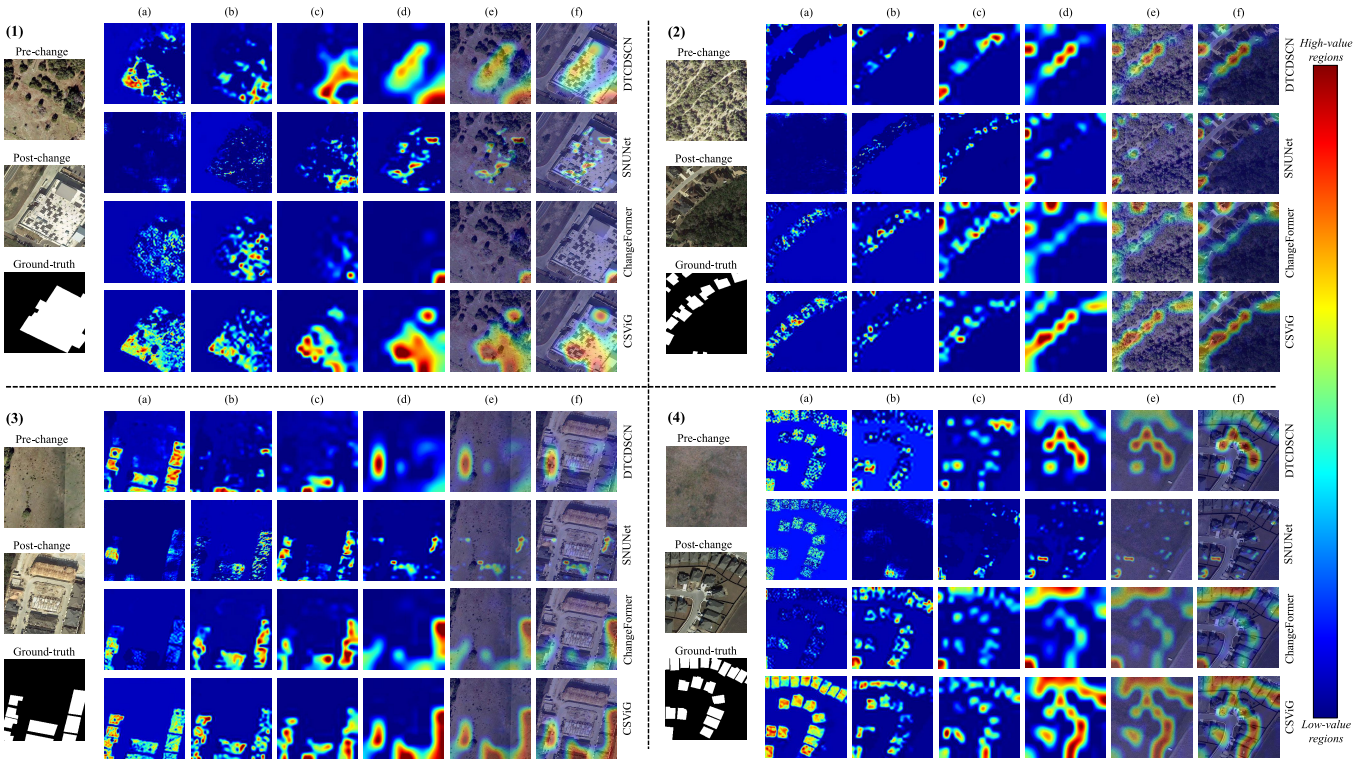


Fig. 9. Visualizations of heat maps at multiple stages for different CD methods. The first column lists the bi-temporal images and corresponding ground-truth from top to bottom. (a)–(d) Results of downsampling (1/4), (1/8), (1/16), and (1/32) feature maps, respectively. (e) Combination of (d) and prechange image. (f) Combination of (d) and postchange image.

TABLE VII

EFFECTIVENESS OF DIFFERENT MODULES WHEN THE MODEL IS TRAINED ON THE LEVIR-CD DATASET. THE VALUES OF EVALUATION METRICS ARE REPORTED AS PERCENTAGE (%)

Methods	SViG	HFFM	VFFM	P	R	$F1$	IoU	OA
(a)	×	×	×	91.12	88.64	89.86	81.59	98.98
(b)	×	✓	✓	91.64	90.23	91.00	83.31	99.08
(c)	✓	×	×	90.77	89.86	89.64	82.82	99.01
(d)	✓	✓	×	91.86	90.52	90.87	83.71	99.08
(e)	✓	×	✓	92.27	90.36	91.13	84.08	99.11
(f)	✓	✓	✓	**92.35**	**90.71**	**91.52**	**84.37**	**99.14**

conducted by visualizing feature maps at different stages to demonstrate that SViG can effectively capture various dependencies. The HFFM obtains differential features by comparing features of pre- and postchange at the same horizontal stage, and the VFFM obtains intermediate change maps by fusing the refined differential features from adjacent vertical stages. We further conduct experiments to qualitatively analyze the contributions of SViG, HFFM, and VFFM. As shown in Table VII, (a) serves as the baseline using ChangFormer [58]. In (b), ResNet50 is utilized as the backbone, while simultaneously embedding HFFM and VFFM. Compared to (b), (f) achieves better performance by using SViG as the backbone when keeping other factors consistent, resulting in an improvement in all metrics. The improvements in these numbers further demonstrate the effectiveness of SViG.

Subsequently, based on the usage of SViG, we train the detection model without embedding HFFM and VFFM on LEVIR-CD, as shown in (c). Next, we separately add HFFM and VFFM into the model to perform training. When only HFFM is used, the $F1$/IoU/OA score is improved by 1.21/0.89/0.07%, and when only VFFM is used, the $F1$/IoU/OA score is improved by 1.49/1.19/0.10%. Finally, we merge them into the model together, and the $F1$/IoU/OA score exceeds method (c) by 1.88/1.55/0.13%. We believe that HFFM effectively detects the difference between changed and unchanged regions, and VFFM improves detection performance by fusing complementary information from multiple stages. It is important to note that VFFM and HFFM cannot be incorporated into certain models of the other methods under consideration. Consequently, our evaluation is limited to comparing the impact of these two modules on SViG.

4) Value of k Nearest Neighbor Nodes: We need to construct edges between each node and its k nearest neighbor nodes when the graph data is built. Therefore, we also conduct ablation experiments to determine the appropriate value for k, and the experimental results are reported in Table VIII. It can be observed from the table that when the value of k is 9, four of the evaluation metrics achieve the highest scores. So, in the other experiments, we fixedly select the nine nearest neighbor nodes of each node in SViG to extract the features.

5) DSConv Value: In HFFM, multiscale DSConv is used to focus on local feature information. We also perform experiments to verify the effect of the number and size of DSConv on the model performance. We choose DSCov kernel size values of 1×1, 3×3, 5×5, and 7×7, respectively, and the experimental results are presented in Table IX. When we fail to apply any DSConv, meaning that we focus only on global context and not on local feature

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TABLE VIII

EXPERIMENTAL RESULTS OF DIFFERENT NUMBER OF NEAREST NEIGHBOR NODES BEING USED IN SViG. THE VALUES OF EVALUATION METRICS ARE REPORTED AS PERCENTAGE (%)

Methods	k	P	R	$F1$	IoU	OA
(a)	6	**92.43**	90.24	91.32	84.02	99.13
(b)	9	92.35	**90.71**	**91.52**	**84.37**	**99.14**
(c)	12	92.34	90.59	91.46	84.25	99.14

TABLE IX

EXPERIMENTAL RESULTS OF MULTISCALE DSCONV WITH DIFFERENT KERNELS BEING USED IN HFFM WHEN THE MODEL IS TRAINED ON THE LEVIR-CD DATASET. THE VALUES OF EVALUATION METRICS ARE REPORTED AS PERCENTAGE (%)

Methods	1×1	3×3	5×5	7×7	P	R	$F1$	IoU	OA
(a)	×	×	×	×	91.34	89.88	90.72	82.22	99.08
(b)	✓	×	×	×	92.02	89.31	90.94	83.42	99.09
(c)	×	×	×	✓	91.92	90.13	91.02	83.52	99.09
(d)	✓	✓	×	×	92.51	90.48	91.49	84.31	99.14
(e)	✓	×	×	✓	**92.55**	90.12	91.32	84.02	99.13
(f)	✓	✓	✓	×	92.35	**90.71**	**91.52**	**84.37**	**99.14**
(g)	×	✓	✓	✓	92.34	90.50	91.40	84.18	99.13

information, the detection performance decreases significantly as shown in (a). As can be seen from (b) to (c), when a single convolution is used, the accuracy increases as the kernel size becomes larger. In addition, experimental data in the table show that the detection performance of the model is improved as the number of convolutions increases. In summary, these experimental data validate that multiscale DSConv enables the model to focus on more local information and improve performance.

6) Data Augmentation: Data augmentation on training samples is one of the most common techniques. In the visual task, data enhancement of images not only increases the diversity of training data, but also improves the robustness of the trained model. Almost all deep-learning-based vision models use data augmentation on the training set, and we also do an interesting experiment about data enhancement as shown in Table X. First, we train the model as a baseline when no data augmentation is available on both the training and test sets as shown in (a). Next, for the test samples without any data augmentation, we just added different data augmentation on the training samples. We can observe that the CD performance is slightly improved when blurring and color jitter are used separately, and the accuracy is further enhanced if they are used together. Therefore, data augmentation improves the robustness of the model to a certain extent. Then, we use data augmentation on the training samples when the model is trained. But during the inference phase on the test set, we add different perturbations to test data as well, and experimental results are presented in (d)–(f). It is clear that the model can achieve the best performance when no perturbations are added to the test samples, as shown in (d). However, when perturbations are also added to the test images, there is only a slight degradation in performance, and this effect is almost negligible. This further demonstrates that data augmentation can improve the robustness and generalization of the model. In addition, by adding Gaussian noise to the training samples in (h), the experimental results on the test set without data enhancement show only a slight decrease from the baseline (a).

TABLE X

COMPARATIVE PERFORMANCE ANALYSIS IN THE PRESENCE OF BLURRING AND COLOR JITTER ON THE TEST SET OF THE LEVIR-CD DATASET. THE VALUES OF EVALUATION METRICS ARE REPORTED AS PERCENTAGE (%)

Methods	Training			Testing			P	R	$F1$	IoU	OA
	blurring	color jitter	noise	blurring	color jitter	noise					
(a)	×	×	×	×	×	×	92.49	90.04	91.25	83.92	99.12
(b)	✓	×	×	×	×	×	**92.69**	89.94	91.30	83.99	99.13
(c)	×	✓	×	×	×	×	91.80	**90.95**	91.37	84.11	99.13
(d)	✓	✓	×	×	×	×	92.35	90.71	**91.52**	**84.37**	**99.14**
(e)	✓	✓	×	✓	×	×	92.37	90.17	91.36	84.08	99.13
(f)	✓	✓	×	×	✓	×	92.02	90.84	91.43	84.22	99.14
(g)	✓	✓	×	✓	✓	×	92.40	90.38	91.37	84.10	99.13
(h)	×	×	✓	×	×	×	91.77	90.57	91.17	83.77	99.11

TABLE XI

TIME COMPLEXITY ANALYSIS OF DIFFERENT CD METHODS

Methods	Params	Train times	Inference times	IoU (%)
FC-EF [18]	1.35 M	9.38 h	12.28 ms	57.70
FC-Siam-Di [18]	1.35 M	9.51 h	17.20 ms	67.16
FC-Siam-Conc [18]	1.55 M	9.75 h	17.29 ms	71.42
DTCDSCN [5]	41.07 M	11.33 h	56.35 ms	80.87
IFNet [30]	35.99 M	25.64 h	27.24 ms	80.36
SNUNet [47]	12.03 M	22.32 h	26.47 ms	81.47
BIT [7]	12.40 M	13.68 h	55.69 ms	81.30
ChangeFormer [58]	41.03 M	58.34 h	85.10 ms	81.59
ICIFNet [63]	25.83 M	41.76 h	152.51 ms	81.75
DMINet [64]	6.76 M	15.35 h	31.23 ms	82.99
CSViG	41.31 M	65.71 h	145.38 ms	**84.37**

B. Time Complexity

While our proposed CD method has significantly improved the performance of segmentation, it is also necessary to conduct a detailed analysis of its parametric quantities and time complexity. All experiments were performed on a single Titan XP GPU, and different models were trained using the same number of epochs. The comparison of experimental results is shown in Table XI. As shown in the table, early CD models [18] using only convolution operations and simple feature fusion way exhibit the fewest parameters and fastest training time, but their accuracy is usually the lowest. Some methods [5], [30], [47] increase computation time because of the use of attention mechanisms, yet the accuracy is correspondingly improved. Other approaches [7], [58] employ Transformer to further improve the segmentation accuracy, although this comes at the expense of longer training times.

Although our proposed method achieves the best CD performance, the training time and inference time of the model are dramatically increased due to SViG reasoning about the dependencies between changed regions of interest and the crossed feature fusion that invokes the attention mechanism and multiscale convolution. However, as Table XI indicates, our proposed method adds only slight model parameters than DTCDSCN [5] and ChangeFormer [58], while exhibiting significantly improved CD performance. These results demonstrate that although the training process and inference process of our method may not be as efficient as some others, its CD performance outperforms current related methods.

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VI. Conclusion

In this work, an effective SViG with crossed feature fusion is proposed to address the existing challenges for the RS image-building CD task. Our model first utilizes a weight-sharing SViG as an encoder that gains short- and long-range dependencies in RS bi-temporal images, capturing adequate global and local information. Then, crossed feature fusion is performed from different orientations. HFFM can easily distinguish real-change regions and suppress pseudo-change objects by using CC, attention mechanisms, and multiscale DSConv to fuse pre- and postchange features in the same horizontal stage. VFFM further fuses complementary content from vertical multiple stages to validly represent the change region. Finally, we demonstrate the effectiveness of our proposed method by performing a series of experiments on three building CD datasets.

Although our method outperforms other current methods in terms of performance, the time it consumes during the training phase increases accordingly. In the future, we will further design new methods that are more lightweight and have better performance. One of our current solutions to the high complexity problem is to use model compression pruning techniques for bulky CD models. Meanwhile, we will explore semisupervised methods suitable for the CD task to overcome the expensive manual annotation problem in RS images, the idea is to use an approach based on consistency regularization and clustering to keep the same categories close and the different categories far apart.

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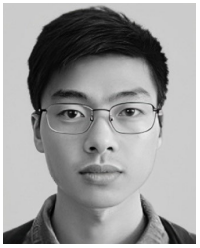
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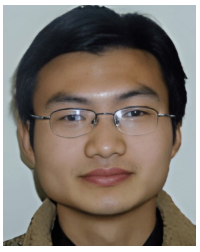
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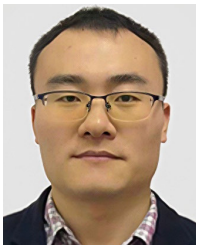
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